Compiler Construction

Lab 6

Submitted By:

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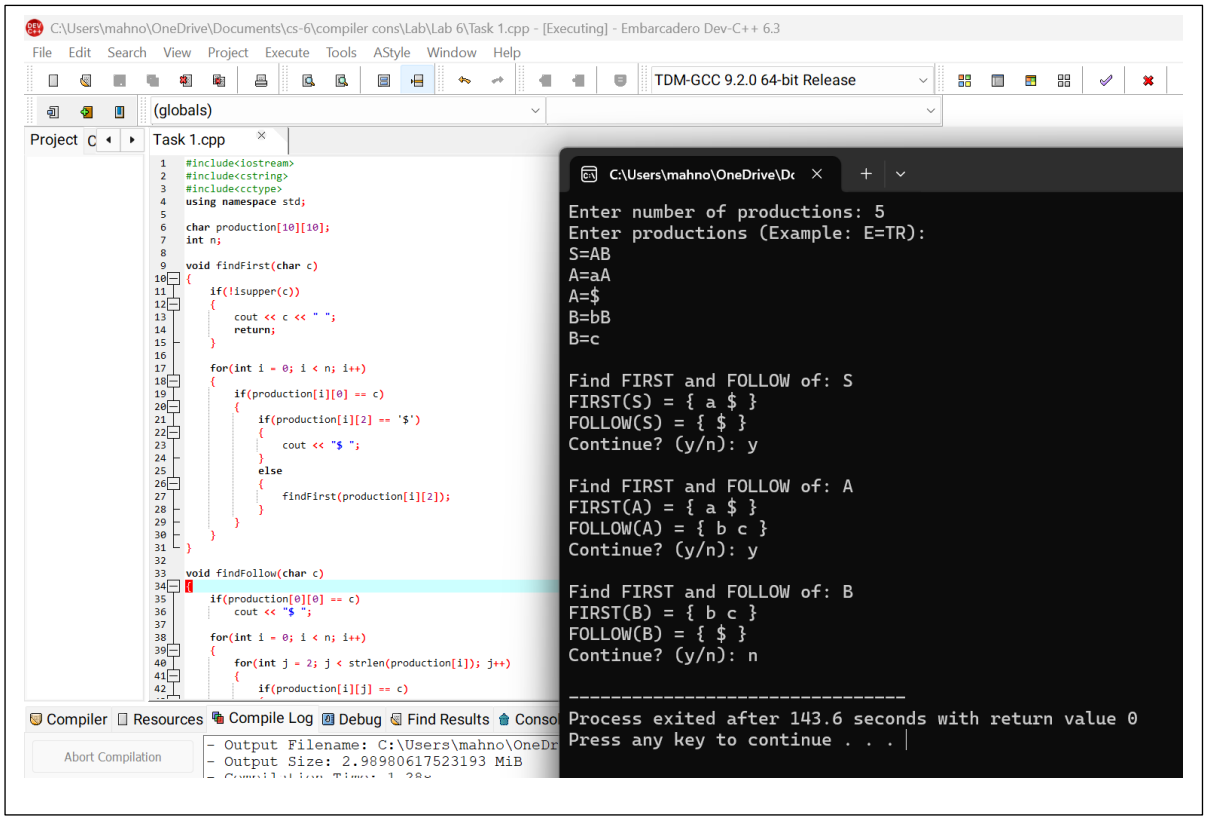
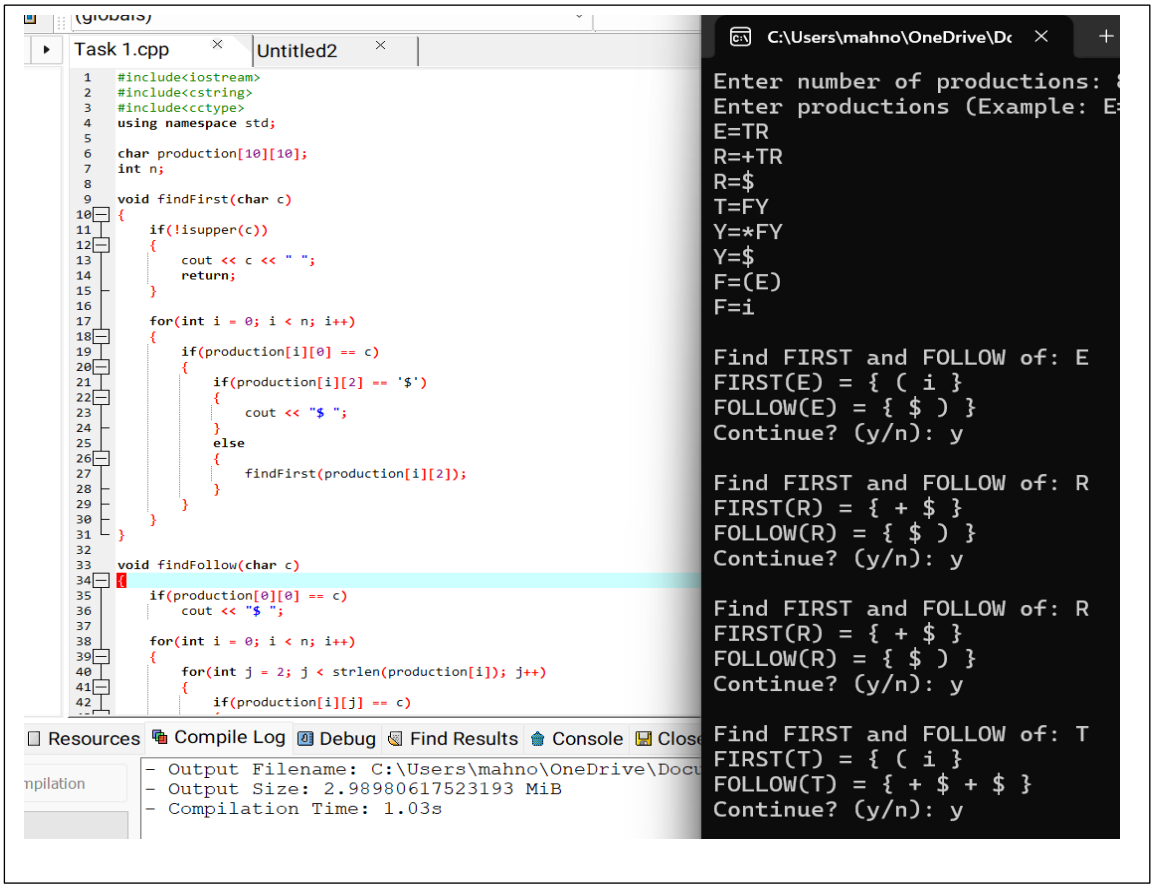
BSCS-6

Task 1:

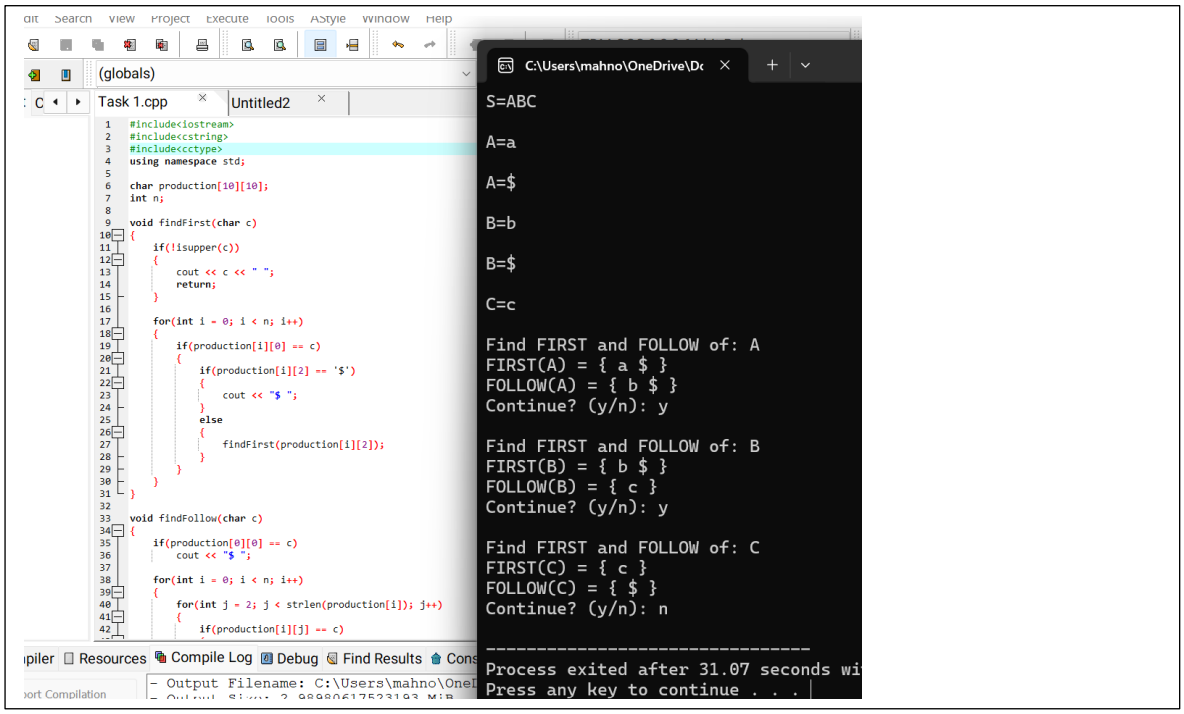



```cpp
                }

                if(production[i][j + 1] == '\0' && c != production[i][0])
                {
                    findFollow(production[i][0]);
                }
            }
        }
    }
}

int main()
{
    cout << "Enter number of productions: ";
    cin >> n;

    cout << "Enter productions (Example: E=TR):\n";
    for(int i = 0; i < n; i++)
        cin >> production[i];

    char choice;
    do
    {
        char c;
        cout << "\nFind FIRST and FOLLOW of: ";
        cin >> c;

        cout << "FIRST(" << c << ") = { ";
        findFirst(c);
        cout << "}\n";

        cout << "FOLLOW(" << c << ") = { ";
        findFollow(c);
        cout << "}\n";

        cout << "Continue? (y/n): ";
        cin >> choice;

    } while(choice == 'y' || choice == 'Y');

    return 0;
}
```

Task 2:


Task 1.cpp
Untitled2
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;
char production[10][10];
int n;
void findFirst(char c)
{
if(!isupper(c))
{
cout << c << " ";
return;
}
for(int i = 0; i < n; i++)
{
if(production[i][0] == c)
{
if(production[i][2] == '$')
{
cout << "$ ";
}
else
{
findFirst(production[i][2]);
}
}
}
}
void findFollow(char c)
{
if(production[0][0] == c)
cout << "$ ";
for(int i = 0; i < n; i++)
{
for(int j = 2; j < strlen(production[i]); j++)
{
if(production[i][j] == c)
Resources
Compile Log
Debug
Find Results
Console
- Output Filename: C:\Users\mahno\OneDrive\Docu
- Output Size: 2.98980617523193 MiB
- Compilation Time: 1.03s
C:\Users\mahno\OneDrive\Dc
Enter number of productions:
Enter productions (Example: E
E=TR
R=+TR
R=$
T=FY
Y=*FY
Y=$
F=(E)
F=i
Find FIRST and FOLLOW of: E
FIRST(E) = { (i }
FOLLOW(E) = { $) }
Continue? (y/n): y
Find FIRST and FOLLOW of: R
FIRST(R) = { + $ }
FOLLOW(R) = { $) }
Continue? (y/n): y
Find FIRST and FOLLOW of: R
FIRST(R) = { + $ }
FOLLOW(R) = { $) }
Continue? (y/n): y
Find FIRST and FOLLOW of: T
FIRST(T) = { (i }
FOLLOW(T) = { + $ + $ }
Continue? (y/n): y

Task 3:


(globals)
Task 1.cpp
Untitled2
Resources
Compile Log
Debug
Find Results
C:\Users\mahno\OneDrive\Dc
S=ABC
A=a
A=$
B=b
B=$
C=c
Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c }
Continue? (y/n): y
Find FIRST and FOLLOW of: C
FIRST(C) = { c }
FOLLOW(C) = { $ }
Continue? (y/n): n
Process exited after 31.07 seconds wi
Press any key to continue . . .

Task 4:

Task 1.cpp | Untitled2

```cpp
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;

char production[10][10];
int n;

void findFirst(char c)
{
    if(!isupper(c))
    {
        cout << c << " ";
        return;
    }

    for(int i = 0; i < n; i++)
    {
        if(production[i][0] == c)
        {
            if(production[i][2] == '$')
            {
                cout << "$ ";
            }
            else
            {
                findFirst(production[i][2]);
            }
        }
    }
}

void findFollow(char c)
{
    if(production[0][0] == c)
        cout << "$ ";

    for(int i = 0; i < n; i++)
    {
        for(int j = 2; j < strlen(production[i]); j++)
        {
            if(production[i][j] == c)
```

C:\Users\mahno\OneDrive\Dc

```
Enter number of productions:
Enter productions (Example: E
S=aS
S=bA
A=cA
A=d

Find FIRST and FOLLOW of: S
FIRST(S) = { a b }
FOLLOW(S) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: A
FIRST(A) = { c d }
FOLLOW(A) = { $ }
Continue? (y/n): n

--------------------------------
Process exited after 14.91 se
Press any key to continue . .
```

Task 5:

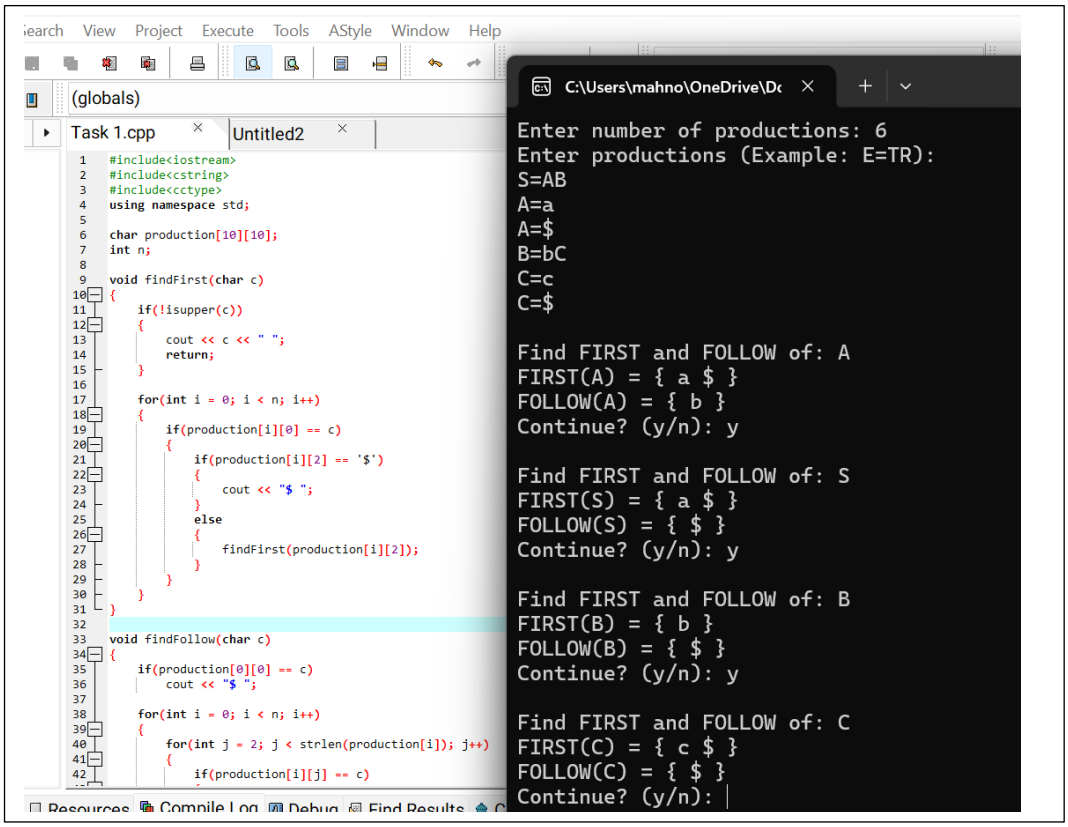
Task 1.cpp | Untitled2

```cpp
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;

char production[10][10];
int n;

void findFirst(char c)
{
    if(!isupper(c))
    {
        cout << c << " ";
        return;
    }

    for(int i = 0; i < n; i++)
    {
        if(production[i][0] == c)
        {
            if(production[i][2] == '$')
            {
                cout << "$ ";
            }
            else
            {
                findFirst(production[i][2]);
            }
        }
    }
}

void findFollow(char c)
{
    if(production[0][0] == c)
        cout << "$ ";

    for(int i = 0; i < n; i++)
    {
        for(int j = 2; j < strlen(production[i]); j++
        {
```

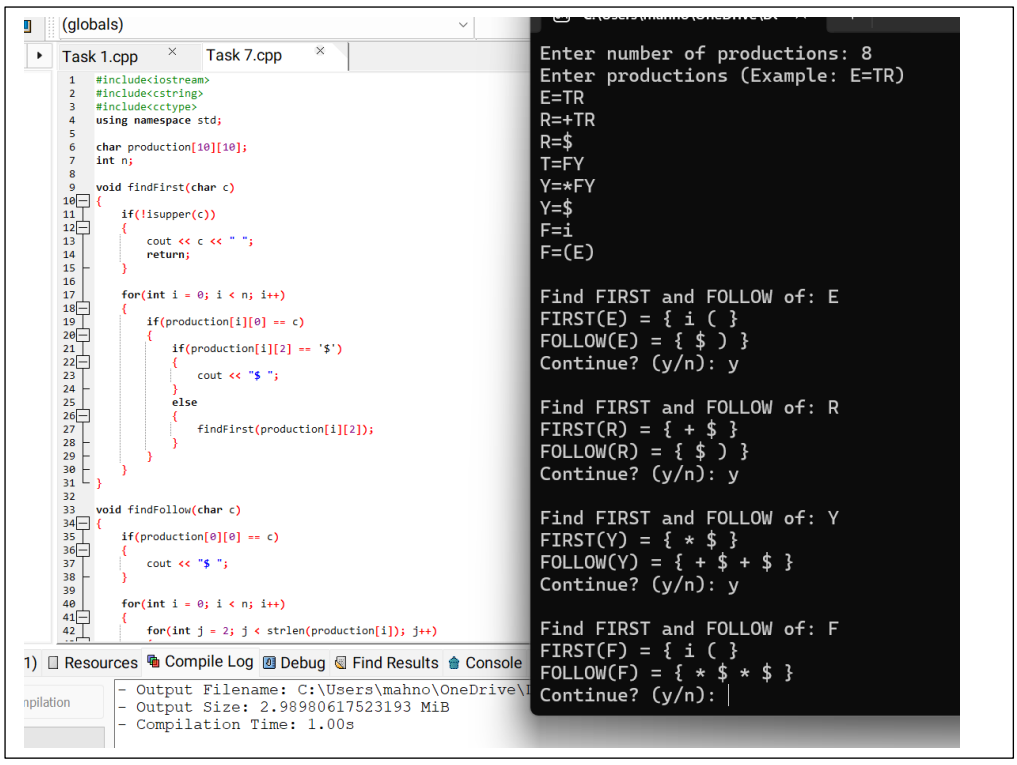
C:\Users\mahno\OneDrive\Dc

```
Enter number of productions: 2
Enter productions (Example: E=TR):

S=(S)S
S=$

Find FIRST and FOLLOW of: S
FIRST(S) = { ( $ }
FOLLOW(S) = { $ ) }
Continue? (y/n): n

--------------------------------
Process exited after 25.92 seconds
Press any key to continue . . .
```

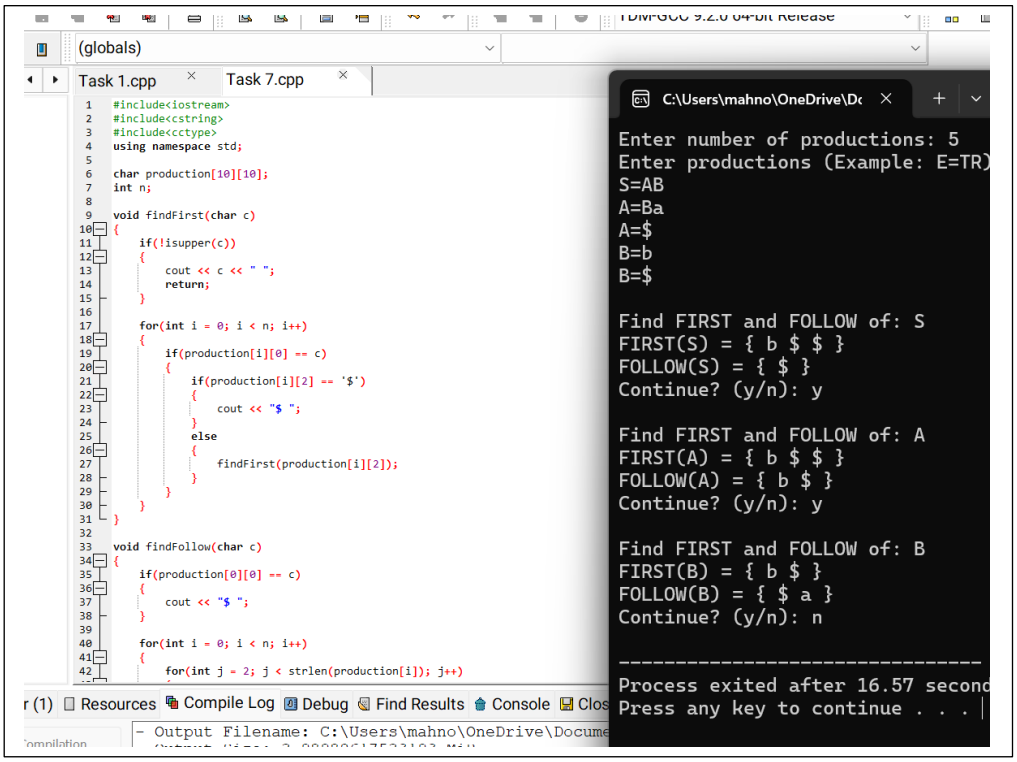
Task 6:


Search View Project Execute Tools AStyle Window Help
(globals)
Task 1.cpp
Untitled2
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;
char production[10][10];
int n;
void findFirst(char c)
{
if(!isupper(c))
{
cout << c << " ";
return;
}
for(int i = 0; i < n; i++)
{
if(production[i][0] == c)
{
if(production[i][2] == '$')
{
cout << "$ ";
}
else
{
findFirst(production[i][2]);
}
}
}
}
void findFollow(char c)
{
if(production[0][0] == c)
cout << "$ ";
for(int i = 0; i < n; i++)
{
for(int j = 2; j < strlen(production[i]); j++)
{
if(production[i][j] == c)
Resources Compile Log Debug Find Results
C:\Users\mahno\OneDrive\Dc
Enter number of productions: 6
Enter productions (Example: E=TR):
S=AB
A=a
A=$
B=bC
C=c
C=$
Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b }
Continue? (y/n): y
Find FIRST and FOLLOW of: S
FIRST(S) = { a $ }
FOLLOW(S) = { $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: B
FIRST(B) = { b }
FOLLOW(B) = { $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: C
FIRST(C) = { c $ }
FOLLOW(C) = { $ }
Continue? (y/n):

Task 7:

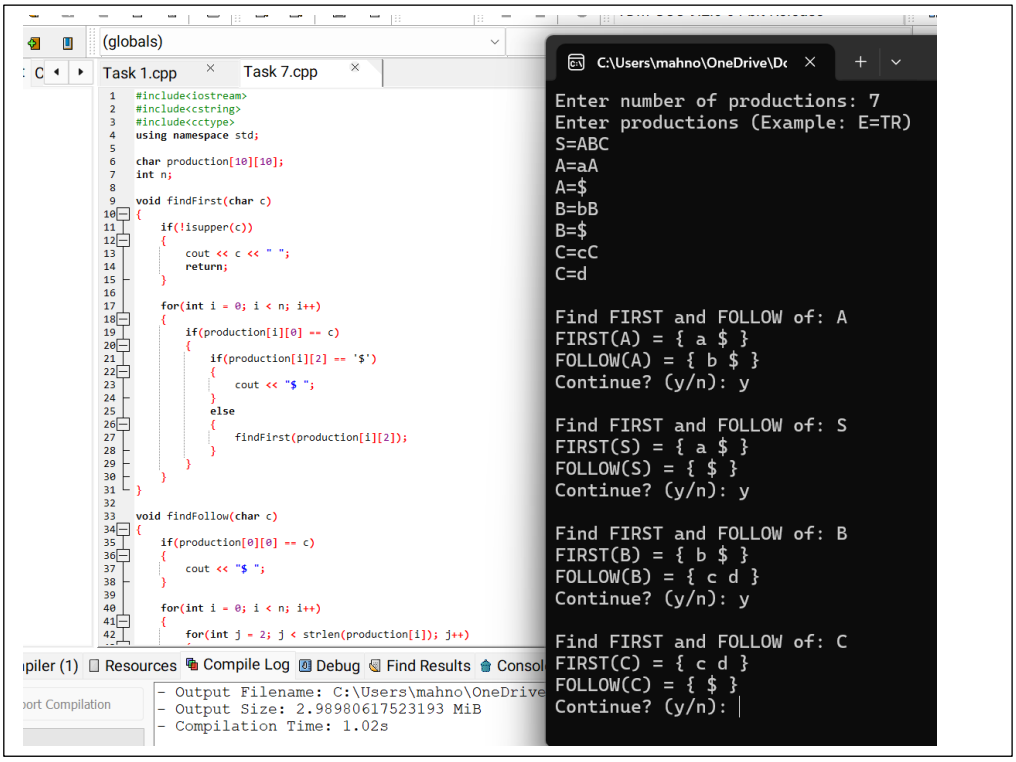

(globals)
Task 1.cpp
Task 7.cpp
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;
char production[10][10];
int n;
void findFirst(char c)
{
if(!isupper(c))
{
cout << c << " ";
return;
}
for(int i = 0; i < n; i++)
{
if(production[i][0] == c)
{
if(production[i][2] == '$')
{
cout << "$ ";
}
else
{
findFirst(production[i][2]);
}
}
}
}
void findFollow(char c)
{
if(production[0][0] == c)
{
cout << "$ ";
}
for(int i = 0; i < n; i++)
{
for(int j = 2; j < strlen(production[i]); j++)
Resources Compile Log Debug Find Results Console
- Output Filename: C:\Users\mahno\OneDrive\
- Output Size: 2.98980617523193 MiB
- Compilation Time: 1.00s
Enter number of productions: 8
Enter productions (Example: E=TR)
E=TR
R=+TR
R=$
T=FY
Y=*FY
Y=$
F=i
F=(E)
Find FIRST and FOLLOW of: E
FIRST(E) = { i (}
FOLLOW(E) = { $) }
Continue? (y/n): y
Find FIRST and FOLLOW of: R
FIRST(R) = { + $ }
FOLLOW(R) = { $) }
Continue? (y/n): y
Find FIRST and FOLLOW of: Y
FIRST(Y) = { * $ }
FOLLOW(Y) = { + $ + $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: F
FIRST(F) = { i (}
FOLLOW(F) = { * $ * $ }
Continue? (y/n):

Task 8:


(globals)
Task 1.cpp
Task 7.cpp
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;
char production[10][10];
int n;
void findFirst(char c)
{
if(!isupper(c))
{
cout << c << " ";
return;
}
for(int i = 0; i < n; i++)
{
if(production[i][0] == c)
{
if(production[i][2] == '$')
{
cout << "$ ";
}
else
{
findFirst(production[i][2]);
}
}
}
}
void findFollow(char c)
{
if(production[0][0] == c)
{
cout << "$ ";
}
for(int i = 0; i < n; i++)
{
for(int j = 2; j < strlen(production[i]); j++)
Resources
Compile Log
Debug
Find Results
Console
Output Filename: C:\Users\mahno\OneDrive\Docume
C:\Users\mahno\OneDrive\Dc
Enter number of productions: 5
Enter productions (Example: E=TR)
S=AB
A=Ba
A=$
B=b
B=$
Find FIRST and FOLLOW of: S
FIRST(S) = { b $ $ }
FOLLOW(S) = { $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: A
FIRST(A) = { b $ $ }
FOLLOW(A) = { b $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { $ a }
Continue? (y/n): n

Process exited after 16.57 second
Press any key to continue . . .

Task 9:


(globals)
Task 1.cpp
Task 7.cpp
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;
char production[10][10];
int n;
void findFirst(char c)
{
if(!isupper(c))
{
cout << c << " ";
return;
}
for(int i = 0; i < n; i++)
{
if(production[i][0] == c)
{
if(production[i][2] == '$')
{
cout << "$ ";
}
else
{
findFirst(production[i][2]);
}
}
}
}
void findFollow(char c)
{
if(production[0][0] == c)
{
cout << "$ ";
}
for(int i = 0; i < n; i++)
{
for(int j = 2; j < strlen(production[i]); j++)
Resources
Compile Log
Debug
Find Results
Output Filename: C:\Users\mahno\OneDrive
Output Size: 2.98980617523193 MiB
Compilation Time: 1.02s
C:\Users\mahno\OneDrive\Dc
Enter number of productions: 7
Enter productions (Example: E=TR)
S=ABC
A=aA
A=$
B=bB
B=$
C=cC
C=d
Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: S
FIRST(S) = { a $ }
FOLLOW(S) = { $ }
Continue? (y/n): y
Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c d }
Continue? (y/n): y
Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { $ }
Continue? (y/n):

Task 10:

Task 1.cpp | Task 7.cpp

```cpp
#include<iostream>
#include<cstring>
#include<cctype>
using namespace std;

char production[10][10];
int n;

void findFirst(char c)
{
    if(!isupper(c))
    {
        cout << c << " ";
        return;
    }

    for(int i = 0; i < n; i++)
    {
        if(production[i][0] == c)
        {
            if(production[i][2] == '$')
            {
                cout << "$ ";
            }
            else
            {
                findFirst(production[i][2]);
            }
        }
    }
}

void findFollow(char c)
{
    if(production[0][0] == c)
    {
        cout << "$ ";
    }

    for(int i = 0; i < n; i++)
    {
        for(int j = 2; j < strlen(production[i]); j++)
```

Resources | Compile Log | Debug | Find Results

```
- Output Filename: C:\Users\mahno\One
- Output Size: 2.98980617523193 MiB
- Compilation Time: 0.98s
```

```
Enter number of productions: 7
Enter productions (Example: E=TR)
S=ACB
A=aA
A=$
B=bB
B=$
C=c
C=d

Find FIRST and FOLLOW of: S
FIRST(S) = { a $ }
FOLLOW(S) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { c d }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { b $ }
Continue? (y/n):
```

Handwritten:

Date

Compiler Construction Lab 6:

1) $S \to AB$
$A \to aA \mid \varepsilon$
$B \to bB \mid c$

First:

~~First(A) = {a, ε}~~

$A \to aA$
$A \to \varepsilon$
First (A) = {a, ε}
First (S) = First(A)
First(S) = {a, b, c}

$B \to bB$
$B \to c$
First(B) = {b, c}
Rule
we check B as well cause ε exists

Follow:

Follow(S) = {$}, Follow(A) = First(B)
Rule = First(B) = {b, c}.
Follow(A) = {b, c}
Follow(B) = {$}

2) $E \to TR$
$R \to +TR \mid \varepsilon$
$T \to FY$
$Y \to *FY \mid \varepsilon$
$F \to (E) \mid id$

$R \to +TR$
$R \to \varepsilon$
First(R) = {+, ε}

Rule
First(E) = First(T)
= {(, id}

First:

~~First(F)~~ $F \to (E)$
$F \to id$
First (F) = {(, id}

$Y \to *FY$
$Y \to \varepsilon$
First(Y) = {*, ε}

Firs(T) = First(F)
Firs(T) = {(, id} Rule

Date

Follow

Follow(E) = {), $ }

Follow(R) {), $ }

Follow(T) = { +,), $ }

Follow(Y) = { +,), $ }

Follow(F) = { *, +,), $ }

3) S → ABC

A → a | ε

B → b | ε

C → c

First:

First(A) = { a, ε }

First(B) = { b, ε }

First(C) = { c }

First(S) = { a, b, c }

Follow:

Follow(S) = { $ }

Follow(A) = { b, c }

Follow(B) = { c }

Follow(C) = { $ }

4) S → aS | bA

A → cA | d

First:

First(S) ⇒ { a, b } Rule: S → aS, S → bA (First Symbols)

First(A) ⇒ { c, d } Rule: A → cA, A → d (First Symbols)

Follow:

Follow(S) = { $ }

Follow(A) = Follow(S) = { $ }

Date

5) $S \to (S)S \mid \varepsilon$

First:

$S \to (S)S$
$S \to \varepsilon$ } Rule : first Symbols

$First(S) = \{(, \varepsilon\}$

Follow:

$Follow(S) = \{), \$\}$

we check what comes after each S.

6) $S \to AB$
$A \to a \mid \varepsilon$
$B \to bC$
$C \to c \mid \varepsilon$

First:

$A \to a$
$A \to \varepsilon$
$first(A) = \{a, \varepsilon\}$
$C \to c$
$C \to \varepsilon$
$first(C) = \{c, \varepsilon\}$
$first(B) = \{b\}$
$first(S) = \{a, b\}$

Follow:

$Follow(S) = \{\$\}$
$Follow(A) = First(B) = \{b\}$
$Follow(B) = \{\$\}$
$Follow(C) = Follow(B) = \{\$\}$

7) $E \to TE'$
$E' \to +TE' \mid \varepsilon$
$T \to FT'$
$T' \to *FT' \mid \varepsilon$
$F \to id \mid (E)$

First:

$First(F) = \{id, (\}$
$First(T') = \{*, \varepsilon\}$
$First(T) = \{id, (\}$
$First(E') = \{+, \varepsilon\}$
$First(E) = \{id, (\}$

Follow:

~~$Follow(E) = \{\$\}$~~
$Follow(E) = \{), \$\}$
$Follow(E') = \{), \$\}$

$Follow(T) = \{+,), \$\}$
$Follow(T') = Follow(T) = \{+,), \$\}$
$Follow(F) = \{*, +,), \$\}$

Date

Task 8:

$S \to AB$

$A \to Ba \mid \varepsilon$

$B \to b \mid \varepsilon$

First :

First(B) = {b, ε}

First(A) = {b, a, ε}

First(S) = {a, b, ε}

Follow :

~~S → A~~ Follow(A) = {b, $}

Follow(B) = {a, $}

Follow(S) = {$}

Task 9:

$S \to ABC$

$A \to aA \mid \varepsilon$

$B \to bB \mid \varepsilon$

$C \to cC \mid d$

First :

First(A) = {a, ε}

First(B) = {b, ε}

First(C) = {c, d}

First(S) = {a, b, c, d}

Follow :

Follow(S) = {$}

Follow(A) = {b, c, d}

Follow(B) = {c, d}

Follow(C) = {$}

Task 10:

$S \to ACB$

$A \to aA \mid \varepsilon$

$B \to bB \mid \varepsilon$

$C \to c \mid d$

First :

First(A) = {a, ε}

First(B) = {b, ε}

First(C) = {c, d}

First(S) = {a, c, d}

Follow :

Follow(S) = {$}

Follow(A) = {c, d}

Follow(C) = {b, $}

Follow(B) = {$}